ARAF

STANDARD

Agentic Risk Architecture Framework

Version 3.0

March 2026

Institute for Autonomous Governance Pty Ltd
ACN 696 112 277

Martin, Carly. Agentic Risk Architecture Framework (ARAF) Standard, Version 3.0. Institute for Autonomous Governance Pty Ltd, 2026.

Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0).

This document supersedes the Governance Coherence Addendum v1.0.1 (GCA v1.0.1), which is withdrawn effective the publication date of this Standard.

Contents

Foreword
Part 1 — Preliminary
1. Scope and Application
2. Normative References
3. Terms and Definitions
Part 2 — Framework Architecture
4. Governing Accountability Test
5. The Six Dimensions
6. Multiplier Logic
7. Governance Benchmark Index (GBI)
8. Governance Coherence Index (GCI)
Part 3 — Certification Architecture
9. Certification Tiers
10. Validity Architecture
11. Assessor Accreditation Standard
Part 4 — Governance of the Standard
12. Standard Governance
13. Version History and Amendment Record
Annexure A — Dimensional Scoring Guide
Annexure B — Relationship to External Frameworks
Annexure C — Citation and Reliance Guidance

Foreword

This document is the Agentic Risk Architecture Framework (ARAF) Standard, Version 3.0, published by the Institute for Autonomous Governance Pty Ltd (ACN 696 112 277) as an open public standard.

ARAF is a governance assessment framework for autonomous system deployments. It addresses the institutional governance gap that technical safety standards do not: whether the organisation deploying an autonomous system has the governance architecture to demonstrate accountability for the decisions that system produces at machine scale, without per-step human authorisation.

Version 3.0 consolidates the full ARAF methodology into a single canonical instrument. It absorbs and supersedes the Governance Coherence Addendum v1.0.1 (GCA v1.0.1), which is withdrawn effective the publication date of this Standard. The Governance Coherence Index (GCI) methodology, previously specified in the GCA, is now contained within Clause 8 of this Standard.

The governing test throughout this Standard — who was responsible, what did they know, what did they do — reflects the accountability standard applied in ASIC v Bekier [2026] FCA 196 and embedded in the director duty of care under section 180(1) of the Corporations Act 2001 (Cth) and equivalent duties in other jurisdictions. ARAF is designed to produce the governance evidence that answers those three questions for autonomous system deployments.

This Standard is the legitimacy foundation for the ARAF certification ecosystem. The companion publication, Trust Architecture: Making Autonomous Systems Bankable (Martin, forthcoming), provides the full analytical framework, worked case studies, and institutional translation for boards, insurers, investors, and regulators.

This Standard is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0). It may be reproduced and built upon with attribution.

Carly Martin *Founder and Director, Institute for Autonomous Governance Pty Ltd*
March 2026

Part 1 — Preliminary

1. Scope and Application

This Standard specifies the Agentic Risk Architecture Framework (ARAF): a methodology for assessing, scoring, and certifying the governance posture of autonomous system deployments.

This Standard applies to any organisation deploying an autonomous system that produces consequential decisions, whether or not those decisions are subject to per-step human authorisation. It is designed for use by:

- accredited ARAF assessors conducting governance assessments;
- boards and governing bodies exercising oversight of autonomous system deployments;
- insurers evaluating governance posture for underwriting purposes;
- institutional investors conducting governance due diligence;
- regulators assessing governance adequacy across supervised populations; and
- enterprise procurement functions establishing governance requirements for AI vendor selection.

Where an organisation operates multiple autonomous decision systems that share infrastructure components (common contract frameworks, shared data pipelines, overlapping vendor relationships, or centralised governance bodies), the shared components are assessed once and the dimensional scores attributed to each system that relies on them. A governance weakness in shared infrastructure affects the dimensional score of every system that depends on it. The assessment report must identify shared infrastructure dependencies and document whether the shared component was assessed independently or attributed from another system's assessment. Shared infrastructure does not reduce the assessment burden; it concentrates governance risk.

This Standard does not address technical safety, model accuracy, or system performance. Those matters are governed by applicable technical standards. This Standard addresses the institutional governance layer: whether the organisation deploying the system can demonstrate accountability for the decisions the system produces.

This Standard is an open public standard licensed under Creative Commons Attribution 4.0 International (CC BY 4.0). Assessors, institutional audiences, regulators, and other users may reference and build upon this Standard without proprietary barriers.

2. Normative References

The following documents are referenced normatively in this Standard. Where a referenced document is updated, the version applicable at the time of assessment governs.

- Corporations Act 2001 (Cth), section 180(1) — director duty of care and diligence
- Privacy Act 1988 (Cth), as amended
- APRA Prudential Standard CPS 230 — Operational Risk Management (2024)
- EU Regulation 2024/1689 — Artificial Intelligence Act
- NIST AI Risk Management Framework (AI RMF 1.0, 2023)
- ISO/IEC 42001:2023 — Artificial Intelligence Management Systems

The following ARAF supplementary specifications are referenced by this Standard and maintained as separate instruments:

- ARAF Evidence Infrastructure Standard (EIS-01)
- ARAF Assessor Accreditation Programme Specification
- ARAF Assessor Guidance Notes (GN series)
  – ARAF Governance Benchmark Index Methodology Specification

3. Terms and Definitions

For the purposes of this Standard, the following terms and definitions apply.

3.1 Agentic Bankability

The condition in which an autonomous system deployment satisfies four institutional prerequisites that enable it to be governed, insured, financed, and relied upon by institutional audiences: (a) identifiability — the system can be classified and described with sufficient precision for institutional analysis; (b) measurability — the system's governance posture can be assessed against a defined standard; (c) governability — the organisation can demonstrate active governance over the system's operation; and (d) insurability — the liability exposure created by the system's autonomous decisions can be identified, bounded, and transferred.

3.2 Autonomous Action Consequences (AE3)

The category of outcomes produced by an autonomous system's decisions without per-step human authorisation. AE3 is the liability category that existing frameworks — professional indemnity, product liability, technology errors and omissions — were not designed to address. It encompasses the consequences of decisions made at machine scale, across machine-scale decision volumes, without individual human sign-off on each decision. Dimension 4 (Liability Architecture) is the primary assessment dimension for AE3 exposure.

3.3 Autonomy Gradient

The spectrum of operational autonomy from systems that execute precisely defined instructions (execution end) to systems that exercise discretionary judgement within broad operational parameters and adapt their own decision logic (adaptive autonomy end). The Autonomy Gradient is assessed as Dimension 1 of the ARAF framework.

3.4 Decision Supply Chain

The sequence of systems, actors, and processes through which a consequential decision is produced, extending from data inputs through autonomous systems and human actors to execution infrastructure. The Decision Supply Chain is the unit of governance accountability in the ARAF framework.

3.5 Distributed Decision Infrastructure (DDI)

The network of human experts, autonomous systems, data infrastructure, and external providers through which an organisation produces consequential decisions, operating across organisational boundaries, employment relationships, and jurisdictional lines simultaneously.

3.6 Four-Link Accountability Chain

The four accountability assignments that must be established for each autonomous system deployment: (a) design accountability — for the decisions made in designing and training the system; (b) deployment accountability — for the decision to deploy the system in a given context; (c) operational accountability — for the ongoing operation and oversight of the system; and (d) outcome accountability — for the consequences of decisions the system produces.

3.7 Governance Benchmark Index (GBI)

The composite governance scoring instrument produced by the ARAF assessment. The GBI translates six-dimensional governance assessment into a comparable institutional signal on a scale of 1.0 (strongest governance posture) to 5.0 (critical governance exposure). Lower scores indicate stronger governance. The GBI anchors the certification tier thresholds defined in Clause 9.

3.8 Governance Coherence Index (GCI)

The operational assessment instrument that measures whether a governance architecture is exercised in practice over the assessment window, not merely documented. The GCI is scored on a scale of 0.0 to 1.0 per dimension, where 1.0 represents full operational coherence. The GCI functions as a modifier on GBI dimensional scores for certification tier determination. See Clause 8.

3.9 Governance Signal

The compressed, independently verified communication of governance posture produced by an ARAF assessment. The Governance Signal allows governance quality to be communicated to institutional audiences without requiring those audiences to conduct the full assessment themselves.

3.10 Reconstructability

The property of a governance record that permits the sequence and basis of consequential decisions to be reconstructed after the fact by an independent assessor, regulator, or court, from evidence produced at the time of governance events rather than assembled retrospectively in response to scrutiny.

3.11 Autonomous Intelligence Oversight Committee (AIOC)

The governance body responsible for maintaining oversight of an organisation's autonomous system portfolio. The AIOC is the institutional mechanism through which the four-link accountability chain is exercised at the organisational level. Governance requirements for the AIOC scale with organisational maturity.

Part 2 — Framework Architecture

4. Governing Accountability Test

The governing accountability test for all ARAF assessments is:

> GOVERNING TEST
> Who was responsible? What did they know? What did they do?

This test reflects the accountability standard applied by courts, regulators, and institutional audiences when assessing whether an organisation's governance of an autonomous system was adequate. It is the standard embedded in the director duty of care under section 180(1) of the Corporations Act 2001 (Cth) and equivalent duties in other jurisdictions, as applied in ASIC v Bekier [2026] FCA 196.

The three questions must be answerable for every link in the Decision Supply Chain, not only the autonomous system at its centre. The chain is only as governable as its least governed link.

An ARAF assessment is designed to produce governance evidence that answers these three questions in verifiable form. A governance architecture that cannot answer them is not a governance architecture. It is a governance claim.

5. The Six Dimensions

ARAF assesses governance posture across six dimensions. Each dimension captures a distinct category of governance risk. Together they produce a composite picture of the governance architecture of an autonomous system deployment.

The six dimensions are designed to answer the governing accountability test for a different category of risk. A dimensional profile without a composite score produces governance information. A composite score without a dimensional profile produces a number. Both are required.

Decision Supply Chain Stage	Governance Requirement	ARAF Dimension
Data Inputs	What data does the system process, and under what legal basis?	D2: Data Sensitivity Exposure

Decision Supply Chain Stage	Governance Requirement	ARAF Dimension
Model Processing	At what level of autonomy does the system operate, and is oversight adequate?	D1: Autonomy Gradient
Decision Formation	How is liability for the system's decisions structured and documented?	D4: Liability Architecture
Execution	Do commercial agreements govern the consequences of autonomous execution?	D3: Contract Infrastructure
Institutional Exposure	Has operational dependency created governance vulnerability?	D5: Commercial Leverage
Chain Lifecycle	Can governance be maintained as the system and its deployment evolve?	D6: Adaptive Stability

5.1 D1: Autonomy Gradient

The Autonomy Gradient is the foundational dimension. It establishes the governance burden that all other dimensions must address. A system operating at high autonomy with weak contract infrastructure faces a qualitatively different risk profile than a system at low autonomy with identical contract weakness.

Dimension 1 assesses five sub-factors:

- Operational autonomy level — the categories of decision the system makes without requiring human authorisation for each individual decision.
- Commitment authority — the maximum financial or operational commitment the system can make autonomously.
- Exception handling — what occurs when the system encounters a decision outside its trained parameters.
- Scope boundary enforcement — how the system's operational scope boundaries are enforced technically and contractually.
- Human oversight adequacy — whether the oversight structure is adequate for the autonomy level at which the system operates.

A governance structure designed for supervised systems applied to autonomous ones is not a defence. It is an exhibit.

5.2 D2: Data Sensitivity Exposure

Data Sensitivity Exposure assesses two distinct risk categories. Operational data sensitivity creates immediate exposure. Training data provenance creates latent exposure embedded at training that surfaces through litigation, regulatory inquiry, and investor diligence, often years later.

Operational data sensitivity is assessed against five categories: personal information; health information; financial information; commercially sensitive information; and data subject to jurisdictional restrictions on cross-border transfer.

Training data provenance is assessed separately across three layers: the foundation model layer (developer representations and warranties); the fine-tuning layer (legal basis, intellectual property rights, consent); and the retrieval-augmented generation layer (copyright exposure, accuracy exposure, data protection exposure at inference time).

Training data provenance risk is backward-compounding: the exposure is created at the moment of training but compounds across the system's entire operational history. Contemporaneous provenance documentation cannot be produced retrospectively. It must be established at the time of training or fine-tuning.

5.3 D3: Contract Infrastructure

Contract infrastructure is the governance layer closest to commercial consequence. Gaps in contract infrastructure become visible at the worst possible time: when an adverse outcome has occurred and the parties are determining who bears the loss.

Dimension 3 assesses five sub-factors:

- Master services agreement maturity — whether customer agreements contain AI-specific provisions.
- Vendor agreement adequacy — whether agreements with AI infrastructure providers, foundation model developers, and data suppliers address the governance obligations those relationships create.
- Data processing agreements — whether agreements are in place and adequate for all categories of data the system processes.
- Liability adequacy — whether liability provisions reflect actual exposure from autonomous operation rather than conventional software exposure.
- Negotiation governance — whether the organisation has a defined process for reviewing and approving deviations from standard contract positions.

5.4 D4: Liability Architecture

Liability Architecture addresses the accountability question that autonomous systems create most directly. It assesses how liability for the system's autonomous decisions is structured, documented, and governed.

The central concept is Autonomous Action Consequences (AE3): the outcomes produced by the system's decisions without per-step human authorisation. This is the category that professional indemnity frameworks, product liability frameworks, and technology errors and omissions frameworks were not designed to address.

Dimension 4 assesses four sub-factors:

- Recognition — whether the organisation has explicitly identified and documented the AE3 category for each material autonomous system.
- Liability cap adequacy — whether caps in commercial agreements are adequate for actual AE3 exposure.
- Carve-out structure — whether carve-outs leave material AE3 exposure ungoverned.
- Insurance coverage — whether the organisation's insurance coverage addresses the consequences of autonomous system decisions.

An insurer assessing AE3 coverage requires the Dimension 4 profile to determine whether the liability architecture is adequate for the autonomy level at which the system operates. Without Dimension 4 governance, AE3 exposure is unbounded, and unbounded exposure cannot be underwritten.

5.5 D5: Commercial Leverage

Commercial Leverage assesses the degree to which the organisation has become operationally dependent on the autonomous system in ways that constrain governance. High commercial leverage does not mean the system is performing poorly. It means the cost of remediating governance weaknesses includes the cost of disrupting commercial relationships that depend on the system's continuous operation.

Dimension 5 assesses four sub-factors:

- Operational dependency — the proportion of a critical function that the system processes.
- Revenue concentration — the degree to which revenue depends on the system's continuous operation.
- Remediation commercial resistance — the commercial cost of governance remediation relative to the organisation's tolerance.
- Lock-in dynamics — contractual and technical barriers to provider substitution or system removal.

5.6 D6: Adaptive Stability

Adaptive Stability assesses whether the governance posture that existed at deployment will be maintained as the system evolves. Autonomous systems change their effective behaviour through model updates, training data changes, scope expansion, and adaptive logic without any deliberate decision by the organisation. A system may have been correctly assessed at deployment and may have drifted outside its governance architecture without anyone noticing.

Dimension 6 assesses three sub-factors:

- Update governance — the processes through which model updates are reviewed, approved, and assessed for governance impact.
- Change triggers — the defined events that require governance reassessment.
- Monitoring architecture — the infrastructure through which governance drift is detected between formal assessment cycles.

6. Multiplier Logic

6.1 Principle

Governance weaknesses compound. The multiplier logic formalises the relationship between dimensional weaknesses that produce systemic exposure when they occur in combination. Individual governance weaknesses are manageable. Compound weaknesses are systemic.

Three canonical multiplier conditions are defined. Each captures a specific pattern of dimensional interaction that produces systemic exposure. No additional multipliers are defined in this version of the Standard. Material findings that arise outside the three canonical multiplier conditions must be identified as material dimensional findings in the assessment report and addressed in the remediation roadmap.

Multiplier	Trigger	Increment	Governance Condition
Systemic Escalation	D1 ≥ 4.0 AND D4 ≥ 4.0	+3.0	High autonomy with inadequate liability architecture. The volume of AE3 consequences is high and liability governance is inadequate to manage them.
Infrastructure Collapse	D3 ≥ 4.0 AND D1 ≥ 3.0	+2.0	Significant autonomy with inadequate contract infrastructure. Liability at the execution boundary has no contractual home.

Multiplier	Trigger	Increment	Governance Condition
Leverage Collapse	D5 ≥ 4.0 AND D4 ≥ 3.0	+2.0	High commercial dependency combined with inadequate liability architecture. The system is structurally resistant to remediation.

6.2 Compound Activation Principle

The compound risk of the dimensional combination, not the mechanical threshold alone, determines multiplier activation. Where a dimensional combination creates systemic exposure and one score falls marginally below a stated threshold, the assessor must evaluate whether the combination produces the governance condition the multiplier is designed to capture.

> **ASSESSOR PRINCIPLE**
> Mechanical thresholds are the default. They are not a ceiling on assessor judgement. An assessment that applies thresholds mechanically without evaluating the governance substance those thresholds are designed to detect has not satisfied the assessment standard.

6.3 D2 and D6 in Multiplier Logic

D2 (Data Sensitivity Exposure) and D6 (Adaptive Stability) do not appear in multiplier triggers. This is not an oversight. D2 exposure is a function of data type, jurisdiction, and regulatory regime; it does not amplify other dimensional weaknesses in the same structural way. D6 is a forward-looking stability indicator; its role is to signal governance maintenance risk, not to compound existing dimensional weaknesses.

High scores in D2 or D6 are material findings in a dimensional assessment and must be addressed in the remediation roadmap. They are not treated as multiplier conditions.

6.4 Remediation Sequencing

Where a multiplier is active, remediating the trigger dimension closest to its threshold first typically produces the greatest composite score movement. The multiplier deactivates when either trigger dimension falls below its threshold. Assessment reports must include a remediation pathway that identifies: (a) which trigger dimension is closer to its threshold; (b) the specific governance actions required to move that dimension below threshold; (c) estimated timeframe; and (d) interim risk management measures.

7. Governance Benchmark Index (GBI)

7.1 Scoring Scale

Each dimension is scored on a scale of 1.0 to 5.0. The scale is inverted: 1.0 represents the strongest governance posture; 5.0 represents critical governance exposure. Higher scores indicate greater governance risk.

Score Range	Interpretation
1.0 — 1.75	Strong governance posture. Supports ARAF Certified designation.
1.76 — 2.50	Minimum institutional threshold met. Supports ARAF Compliant designation.
2.51 — 3.50	Material governance weakness requiring remediation.
3.51 — 5.00	High governance risk. Institutional inadequacy. Remediation required before certification.

7.2 Composite Score

The GBI composite score is derived from the six-dimensional assessment and adjusted by multiplier logic where compound weaknesses are present. It is not a simple average of the six dimensional scores.

The composite score answers a narrow but important question: what is the overall governance posture of this deployment in institutionally comparable form? The dimensional profile answers the broader governance question: where, specifically, does the exposure sit? Both outputs are required. Neither substitutes for the other.

A GBI of 3.42 produced by two active multipliers on a system where three dimensions score at or above 3.5 is a fundamentally different governance condition from a GBI of 3.42 produced by six dimensions each scoring approximately 3.4 with no multipliers active. The composite score is the same. The governance problem is not.

The GBI composite score is calculated using the weighting methodology published in the ARAF Governance Benchmark Index Methodology Specification, maintained as a supplementary specification to this Standard by the Institute for Autonomous Governance Pty Ltd. The weighting methodology specifies the relative contribution of each dimension to the composite score and the mechanics by which multiplier increments are applied. Assessors must apply the current published version of the weighting methodology. The weighting methodology is subject to calibration review under Clause 8.7.

7.3 Evidence Quality Tiers

The GBI score communicates governance posture. The evidence tier communicates the confidence warranted in that assessment. All assessment reports must identify the evidence tier used per dimension.

Tier	Evidence Type	Confidence Level
Tier 1: Infrastructure-Generated	Contemporaneous, tamper-evident records produced by governance infrastructure as a natural output of operations.	Highest. Cannot be retrospectively assembled.
Tier 2: Contemporaneous Documentation	Records produced at the time of governance decisions: board papers, AIOC minutes, deployment approval records.	High. Depends on documentation consistency and completeness.
Tier 3: Reconstructed Documentation	Records assembled retrospectively in response to assessment requests.	Lower. Subject to selection and framing choices. May limit certification tier achievable.

7.4 What the GBI Is and Is Not

The GBI is a governance classification tool, a comparability mechanism, a certification input, and an institutional signal. It is not a measure of model accuracy, technical safety, or commercial success. It does not substitute for board judgement or replace the full dimensional assessment.

Technical safety standards and the GBI serve different institutional functions. Technical safety asks whether the system operates correctly. The GBI asks whether the institution deploying it can demonstrate accountability for the decisions the system produces. Both are necessary. Neither substitutes for the other.

8. Governance Coherence Index (GCI)

This clause absorbs and supersedes the ARAF Governance Coherence Addendum v1.0.1. All references to the GCA v1.0.1 in prior assessments and documents should be read as references to this clause of ARAF Standard Version 3.0.

8.1 Purpose

Governance Coherence addresses the most significant limitation of design-only assessment: the decoupling between documented governance architecture and actual operational practice. An organisation may maintain governance documentation of excellent quality while governance controls are not exercised in operation.

Governance Coherence is not a seventh ARAF dimension. It is an evaluation layer applied across all six existing dimensions. The GBI evaluates whether governance architecture is appropriately structured. The GCI evaluates whether that architecture is actually exercised. The GBI and GCI together determine certification tier eligibility for ARAF Compliant and ARAF Certified designations.

The SOC 2 parallel: the GBI assessment corresponds to Type I (point-in-time design adequacy); the GCI assessment corresponds to Type II (operational effectiveness over a defined period).

The GCI does not apply to the ARAF Assessed tier, which remains a design adequacy assessment only.

8.2 Coherence Components

Three components are assessed:

8.2.1 Authority Adherence

Authority Adherence measures whether operational decisions occurred within the authority boundaries defined in the governance architecture. Assessment indicators include: decision logs showing authority level applied at each consequential action; escalation records demonstrating that decisions exceeding defined autonomy thresholds were escalated to the appropriate authority level; and override logs documenting instances where automated decisions were superseded by human intervention.

A pattern of systematic boundary violations — decisions consistently operating at autonomy levels above those defined in the governance architecture — is a critical coherence finding requiring immediate remediation before certification can be issued or maintained.

8.2.2 Control Exercise

Control Exercise measures whether specified governance controls were performed at the required cadence and depth during the assessment window. Assessment indicators include: governance meeting records with timestamps, attendance, and decision records; approval documentation for system changes and autonomy level modifications; monitoring reports demonstrating surveillance obligations were fulfilled; and incident records showing governance processes were triggered by operational events.

The absence of recorded exceptions is a negative coherence indicator. Autonomous system deployments always produce edge cases, boundary conditions, and governance events. Zero exceptions indicates either that the governance process is not being exercised or that exception recording is not functional. Assessors must evaluate the absence of exceptions explicitly.

8.2.3 Drift Detection

Drift Detection measures whether the system's operational behaviour has diverged from the governance assumptions made at the time of the initial or most recent assessment. Drift is specific to autonomous systems: a system may

have been correctly assessed at deployment and may have drifted outside its governance architecture without any deliberate organisational decision.

Assessment indicators include: model version tracking with change impact assessments; operational scope documentation comparing deployed scope against assessed scope; decision pattern analysis identifying systematic changes in decision distributions; and provider change records where third-party AI components have been substituted or updated.

Drift detection is the component that distinguishes Governance Coherence assessment from standard audit methodology and is the primary justification for periodic coherence reassessment rather than one-time certification.

8.3 GCI Scoring Methodology

The GCI is scored on a 0.0 to 1.0 scale, calculated per ARAF dimension. A GCI of 1.0 represents full operational coherence: all governance controls exercised at the required frequency and depth, no authority boundary violations, and no measurable drift from governance assumptions.

The GCI is not a single aggregate score. Dimensional scores must be reported individually to preserve diagnostic utility. A combined GCI is calculated as a weighted average for certification tier determination, with weighting applied based on the system's risk profile as determined by the GBI assessment.

During the calibration period defined in Clause 8.7, the GCI weighting methodology is as follows: each dimension receives equal base weighting (1/6), adjusted by a risk-priority modifier that doubles the weighting of any dimension scoring at or above 3.0 on the GBI and proportionally reduces the weighting of remaining dimensions. This provisional methodology ensures that coherence assessment concentrates scrutiny on the dimensions where governance architecture is weakest. The weighting methodology is subject to review at the calibration checkpoints defined in Clause 8.7 (20 and 50 assessments) and may be revised through the amendment protocol in Clause 12.4.

8.4 GCI as GBI Modifier

The GCI functions as a modifier on the base GBI dimensional scores. For a given dimension, the effective score used for certification tier determination is:

PENALTY FORMULA

Effective Dimension Score = GBI Dimension Score × (2 − GCI Dimension Score) This formula penalises low GCI scores by increasing the effective dimension score. An organisation with strong governance design (low GBI score) but poor operational coherence (GCI below 1.0) will have its effective certification posture degraded to reflect operational reality.

8.5 Evidence Requirements by Tier

Certification Tier	Evidence Requirement
ARAF Assessed	No coherence evidence required. Design adequacy assessment only.
ARAF Compliant	Tier 1 or Tier 2 evidence required. Where Tier 2 is used for D1, D2, or D6, the assessment report must note the limitation. Tier 2 is the expected standard for D3, D4, and D5.
ARAF Certified	Tier 1 evidence required for at least 80% of sampled controls. Tier 2 acceptable for D3, D4, and D5. Any control assessed on Tier 2 evidence must be individually identified in the assessment report.

8.6 Assessment Window

The minimum assessment window for GCI evaluation is 180 days. The assessment window must be specified in the certification report. Assessors must sample governance artefacts across the entire assessment window, not only from periods immediately preceding assessment. Sampling must assess presence, timeliness, and completeness of evidence.

8.7 Calibration Protocol

During the initial deployment period (first 20 certifications), the following calibration requirements apply: dual-assessor review (all GCI scores must be independently validated by a second accredited assessor); range reporting (GCI scores reported as a range rather than a point estimate); inter-rater reliability tracking (scoring differences between assessors must be documented and submitted to the Institute); and threshold review (provisional thresholds reviewed after 20 and 50 assessments).

8.8 Data Handling

Evidence reviewed during coherence assessment must be held under confidentiality obligations equivalent to those applicable to financial audit work papers. Evidence must not be retained beyond the period required for assessment completion and regulatory compliance. Evidence destruction protocols must be specified in the assessment engagement terms. Where evidence contains personal data, assessment must comply with applicable privacy legislation.

Part 3 — Certification Architecture

9. Certification Tiers

9.1 Overview

ARAF defines three certification tiers. Certification is the mechanism that produces the Governance Signal: the compressed, independently verified communication of governance posture that allows governance quality to travel through institutional markets.

Certification performs three functions analogous to credit ratings in bond markets: information compression (translating six-dimensional assessment into a comparable signal); accountability transfer (an accredited assessor accepts professional responsibility for the assessment that produced the certification); and market creation (certification makes governance posture comparable across deployments, enabling pricing of governance risk).

Tier	Threshold	Institutional Meaning
ARAF Assessed	Assessment completed; no minimum score required	Governance posture has been independently evaluated and is available for institutional review. Minimum designation for institutional engagement discussions. Supports a 60-day Reliance Freshness period only.
ARAF Compliant	GBI ≤ 2.50 GCI ≥ 0.70	Minimum institutional threshold met. Governance posture adequate for institutional reliance in most deployment contexts. Supports standard insurance coverage terms and enterprise procurement requirements. 12-month certification validity.
ARAF Certified	GBI ≤ 1.75 GCI ≥ 0.85 Tier 1 evidence ≥ 80% of sampled controls	Full agentic bankability conditions met. Governance posture supports classification, insurance, financing, and institutional reliance. Supports the most favourable coverage terms and the strongest institutional governance signal. 24-month certification validity with mandatory 12-month interim surveillance.

9.2 Initial Assessment

The initial ARAF assessment evaluates governance posture across all six dimensions and produces: GBI composite score; full dimensional profile; GCI

scores (Compliant and Certified tiers); multiplier analysis; evidence tier summary; remediation roadmap where required; certification designation; and assessment date and validity period.

9.3 Self-Certification Prohibition

Self-certification does not satisfy the ARAF independence requirement. An organisation that produces its own ARAF assessment has produced a self-assessment, not a certification. Self-reported governance is evidence of intent. Independent assessment is evidence of fact.

9.4 Certification and the Foreseeability Standard

An organisation that has obtained ARAF Certified or ARAF Compliant status from an accredited assessor holds contemporaneous, independently produced evidence that its governance architecture was assessed against a defined standard and found adequate at the time of assessment.

That evidence does not constitute a statutory safe harbour. It is the kind of contemporaneous evidence that courts assessing the standard of care under section 180(1) of the Corporations Act, insurers evaluating coverage, and regulators conducting supervisory review will find relevant to whether reasonable precautions were taken.

9.5 Certification Withdrawal

Certification may be withdrawn where: material changes were not notified or addressed within the required notification period; governance posture deteriorates below the certified threshold; or the assessment was based on materially inaccurate or incomplete information.

Continued use of a certification after a withdrawal event constitutes a misrepresentation to institutional audiences that rely on it.

10. Validity Architecture

10.1 Canonical Timing Parameters

The following validity parameters are canonical for this version of the Standard and may not be modified by assessors or organisations:

Parameter	Period	Description
Assessment Window (GCI)	180 days minimum	The minimum period over which GCI evidence must be sampled.
Reliance Freshness	60 days from report issue	The period within which an assessment report may be relied upon without

Parameter	Period	Description
		reassessment. Applies to all certification tiers.
Trigger Notification	60 days from triggering event	The period within which an organisation must notify its accredited assessor of a triggering event requiring reassessment.
Scheduled Reassessment (Compliant)	12 months maximum	Maximum interval between formal reassessments for ARAF Compliant designated organisations.
Certification Validity (Compliant)	12 months	Certification validity period for ARAF Compliant designation.
Certification Validity (Certified)	24 months	Certification validity period for ARAF Certified designation.
Interim Surveillance (Certified)	12 months from initial certification	Mandatory interim surveillance assessment for ARAF Certified organisations at 12 months.

10.2 Event-Triggered Reassessment

The following events require mandatory reassessment regardless of the ordinary reassessment cycle:

- Significant model updates, including retraining on materially different data or changes to core model architecture.
- New deployment contexts not covered by the scope of the initial assessment.
- Material changes to data sources, including provider substitution or scope expansion of data inputs.
- Anomaly patterns suggesting governance scope drift identified outside the normal assessment cycle.
- Regulatory developments materially affecting the organisation's governance obligations.
- Adverse outcomes implicating governance architecture.
- Material changes to contract infrastructure, including renegotiation of AI-specific liability provisions.
- Increase in autonomy level of more than one Autonomy Gradient level.

Organisations must notify their accredited assessor of triggering events within 60 days of occurrence. The Reliance Freshness period for the existing

certification does not extend beyond 60 days following a triggering event, regardless of when the event is notified.

11. Assessor Accreditation Standard

11.1 Accreditation Requirements

Certification is only as credible as the assessors who produce it. Accredited ARAF assessors must satisfy four requirements:

- Demonstrated qualification — competence in governance assessment methodology, including the six ARAF dimensions, multiplier logic, GCI methodology, and the evidentiary standard that governance records must satisfy.
- Documented methodology adherence — assessments must be conducted according to the published ARAF methodology. The methodology is open (CC BY 4.0), making the assessment process independently reviewable by any institutional audience that relies on the output.
- Independence — assessors must be independent of the organisation being assessed. No assessor may certify an organisation with which it has a commercial relationship that could compromise assessment objectivity. Advisory organisations may hold assessor accreditation provided advisory and assessment functions are structurally separated.
- Accountability — assessors are subject to quality assurance review. Accreditation may be withdrawn where assessors fail to maintain the assessment standard.

11.2 GCI Assessor Competence

Governance Coherence assessment requires competencies distinct from design adequacy assessment. Accredited assessors must complete the Governance Coherence module of the ARAF Assessor Accreditation Programme before conducting GCI assessments at the Compliant or Certified tier. The module covers: evidence evaluation methodology for each coherence component; sampling design for governance artefact review; drift analysis techniques; GCI scoring mechanics and calibration protocols; and assessment report structure and scope limitation requirements.

11.3 Accreditation Architecture

The accreditation architecture was designed with specific reference to the failure mode documented in the FTC's 2014 action against TRUSTe, where inadequate assessor oversight produced over 1,000 certifications that did not reflect actual practice. ARAF addresses that failure directly: assessors are subject to accountability review, and accreditation is contingent on maintaining assessment quality.

The Assessor Accreditation Programme Specification is maintained as a supplementary specification to this Standard by the Institute for Autonomous Governance Pty Ltd.

11.4 Assessment Report Structure

All ARAF Compliant and Certified assessment reports must include the following sections: Executive Summary (certification tier, GBI score, GCI scores, multiplier status, scope); Scope and Assessment Window; Design Adequacy Assessment (GBI dimensional scores and findings); Governance Coherence Assessment (GCI dimensional scores and findings); Evidence Summary (evidence tier used per dimension, gaps noted); Findings and Observations (critical, significant, informational); Remediation Roadmap; and Scope Limitation and Disclaimer.

The following scope limitation language must appear in every Compliant and Certified assessment report:

REQUIRED SCOPE LIMITATION LANGUAGE

This certification covers the governance architecture design and operational coherence of [Organisation] during the assessment window specified above. It does not guarantee future system performance, safety, or absence of adverse outcomes. Certification is subject to reassessment upon the occurrence of material change events as defined in the ARAF Standard Version 3.0.

Part 4 — Governance of the Standard

12. Standard Governance

12.1 Issuing Body

This Standard is maintained by the Institute for Autonomous Governance Pty Ltd (ACN 696 112 277), the standards body established to hold the intellectual property and governance authority for the ARAF framework.

Governance authority for the published Standard sits with the Institute. Framework stewardship, publication discipline, version control, and interpretive guidance are managed by the Institute under the governance model described in this clause.

12.2 Current Governance Model

The Standard is maintained under a founder-led governance model at publication of Version 3.0. This structure preserves clarity of authorship while establishing the Standard in a stable, citable form that institutional audiences can reference.

The governance model is designed to scale with the ecosystem it supports. The staged maturity trajectory is:

- Current stage (founder-led): governance authority, publication discipline, and interpretive guidance managed by the founder and the Institute. Public versioning and version history maintained. Open licence preserves independent reviewability.
- Ecosystem development stage: technical contributors and certification partners contribute methodology feedback. An institutional advisory group provides input on major revisions. Formal amendment and consultation processes established for material methodology changes.
- Institutional-scale stage: multi-stakeholder governance body with defined composition, meeting cadence, and public reporting. Amendment processes governed by consultation requirements and institutional input.

Transition between stages is governed by ecosystem development milestones rather than fixed timelines. The objective at each stage is transparency of stewardship, not governance complexity.

12.3 Governance Principles

The maintenance of this Standard is governed by five principles:

1. Version discipline. Each public version of this Standard is clearly identified by version number, publication date, and changelog entry. Institutional audiences citing this Standard in procurement requirements, regulatory guidance, or insurance underwriting

guidelines must be able to reference a specific version with confidence that its content is stable.

2. Public transparency. The current version, governance status, amendment history, and citation format are visible to all users of the framework. This Standard is not maintained behind access restrictions.
3. Methodological consistency. Changes to the framework must preserve the internal coherence of the methodology. Dimensional definitions, scoring logic, multiplier thresholds, and certification tier boundaries must not be altered informally or ambiguously. Material changes follow the amendment process defined in Clause 12.4.
4. Institutional usability. This Standard must remain usable by the institutional audiences for whom it is designed: boards, regulators, insurers, investors, enterprise procurement teams, and accredited assessors. Changes that would impair comparability across assessment periods or break backward compatibility with prior certification decisions require explicit justification and transition guidance.
5. Independence of assessment from standard maintenance. The Institute does not hold a privileged position in the assessment or certification ecosystem. Assessor accreditation, certification issuance, and quality assurance operate under defined independence requirements that apply equally to all participants, including the standard maintainer.

12.4 Amendment Protocol

Amendments to this Standard are classified as:

- Major amendments — changes to dimensional definitions, scoring logic, multiplier thresholds, certification tier boundaries, or validity architecture parameters. Major amendments require a 60-day public consultation period before adoption and a formal version increment (e.g., 3.0 to 4.0).
- Minor amendments — corrections, clarifications, and additions that do not affect the scoring architecture or certification thresholds. Minor amendments are published with a minor version increment (e.g., 3.0 to 3.1) and do not require public consultation.
- Interpretive guidance — guidance on the application of existing provisions issued as numbered Governance Notes (GN series). Interpretive guidance does not constitute an amendment and does not require a version increment.

All amendments and interpretive guidance are published in the Version History maintained under Clause 13.

12.5 Scope of Standard Governance

Standard governance covers: version publication and maintenance; methodology updates and clarifications; interpretive guidance for assessors (issued as Governance Notes); governance notes for future revisions; and

stewardship of the published specification, dimensional guidance, GBI methodology, and certification tier definitions.

Standard governance does not constitute: assessor accreditation (governed by the Assessor Accreditation Programme Specification); certification issuance (conducted by accredited assessors); evidence infrastructure certification (governed by EIS-01); or any statutory or regulatory authority.

12.6 Relationship to the Assessor Ecosystem

The Standard defines the methodology. The certification ecosystem operates on top of the published Standard: accredited assessors conduct assessments against the published methodology, issue certifications based on the GBI score, GCI scores, and dimensional profile, and maintain independence from the organisations they assess.

When standard governance produces a material methodology change, that change flows through to the certification ecosystem. Certified organisations are subject to the reassessment obligations described in Clause 10. Assessors are responsible for applying the current published version of the methodology in all assessments.

13. Version History and Amendment Record

Version	Date	Summary
1.0	2025	Initial publication. Six dimensions, GBI scoring, three certification tiers.
2.0	2025	Addition of multiplier logic (Systemic Escalation, Infrastructure Collapse, Leverage Collapse). Dimensional sub-factor specifications.
2.1	2025	GCA v1.0 published as companion document. GCI methodology introduced.
2.1.1	Early 2026	Minor corrections. GCA v1.0.1 issued. Trigger notification corrected to 60 days (from 30 days in GCA v1.0).
3.0	March 2026	Consolidated canonical release. GCI methodology absorbed from GCA v1.0.1 (now superseded). Political Cascade removed as multiplier condition; treated as material finding category in assessor guidance only. Canonical timing parameters locked. Entity updated to Institute for Autonomous Governance Pty Ltd (ACN 696 112 277). Penalty formula for GCI modifier clarified.

Annexure A — Dimensional Scoring Guide

This annexure is informative. It provides assessor reference guidance for sub-factor scoring and compound activation examples. It does not form part of the normative methodology.

A.1 Scoring Orientation

Each dimension is scored 1.0–5.0. The following anchor descriptions guide scoring calibration across the range:

Score	Governance Condition
1.0	Governance architecture is comprehensive, documented, independently verified, and actively exercised. Evidence is Tier 1 across all sub-factors.
2.0	Governance architecture is adequate and documented. Material sub-factors are addressed. Evidence is Tier 1 or Tier 2. Minor gaps do not create systemic exposure.
3.0	Governance architecture is partially adequate. Material gaps exist in one or more sub-factors. Evidence quality is mixed. Remediation is required but not urgent.
4.0	Governance architecture has significant gaps. One or more sub-factors are inadequately addressed. Evidence quality is predominantly Tier 3. Multiplier activation should be evaluated.
5.0	Governance architecture is absent or critically inadequate. The sub-factors required to manage risk in this dimension are not present. Immediate remediation required.

A.2 Compound Activation Example

FinCo Credit Engine (illustrative). D1 score: 3.5 (below the 4.0 Systemic Escalation threshold). D4 score: 4.5 (above threshold). The system executes autonomous credit decisions with AUD 250,000 commitment authority, no per-decision oversight, no AE3 recognition, and a phantom human review mechanism in the recourse process.

Assessor finding: Systemic Escalation active despite marginal D1 shortfall. The combination — high-velocity AE3 generation with no liability structure — produces the governance condition the multiplier is designed to capture. The mechanical threshold is the default; it is not a ceiling on assessor judgement.

A.3 D2 Scoring Example: Training Data Provenance

HealthCo Clinical Decision Support (illustrative). D2 score: 3.5. The system processes patient health records (operational data sensitivity: high) and was fine-tuned on a dataset assembled by a third-party data broker. The deploying organisation holds the data broker's contractual warranty that the dataset was "lawfully obtained" but has not independently verified consent basis, jurisdictional restrictions, or intellectual property clearances for the training corpus. The foundation model layer relies on the developer's standard representations. No RAG-layer provenance documentation exists.

Assessor finding: Operational data sensitivity is adequately governed (health information classification documented, access controls in place, jurisdictional restrictions mapped). Training data provenance is inadequately governed at two of three layers: the fine-tuning layer relies on an unverified third-party warranty rather than documented provenance, and the RAG layer has no provenance documentation. The foundation model layer is assessed on the basis of developer representations, which is the expected standard for that layer. The composite D2 score of 3.5 reflects strong operational governance offset by material provenance gaps at the fine-tuning and RAG layers.

This example illustrates the D2 two-category assessment structure: operational data sensitivity and training data provenance are assessed separately because they create different risk categories with different temporal profiles. Operational exposure is immediate. Provenance exposure is backward-compounding.

Annexure B — Relationship to External Frameworks

This annexure is informative. It describes the relationship between ARAF and external governance, risk, and AI standards frameworks. ARAF occupies the institutional governance certification layer. It is positioned above technical enforcement infrastructure and below statutory regulatory requirements.

Framework	Relationship to ARAF
EU AI Act (2024/1689)	ARAF Dimension 1 maps to the high-risk AI system obligations. ARAF certification provides contemporaneous governance evidence relevant to Article 9 risk management and Article 17 quality management obligations.
NIST AI RMF 1.0	ARAF GBI dimensions map across the four NIST AI RMF functions: Govern (D3, D4, D6), Map (D1, D2), Measure (GBI/GCI), Manage (D5, D6). ARAF provides the institutional certification layer that NIST AI RMF identifies but does not specify.
ISO/IEC 42001:2023	ISO 42001 specifies AI management system requirements. ARAF assesses governance posture of specific autonomous deployments against an outcome-focused institutional standard rather than a management system design standard. The two frameworks are complementary.
APRA CPS 230	CPS 230 Operational Risk Management requires boards to maintain oversight of material service providers and operational risk. D3 and D5 directly address the vendor dependency and contract infrastructure obligations CPS 230 imposes on APRA-regulated entities.
Corporations Act s.180(1)	The governing accountability test (who was responsible, what did they know, what did they do) reflects the section 180(1) duty standard. ARAF certification provides the contemporaneous, independently produced evidence relevant to whether reasonable precautions were taken.
SOC 2 Type I / Type II	The GBI/GCI distinction mirrors the SOC 2 Type I (design adequacy) / Type II (operational effectiveness) distinction. ARAF is designed to function as the autonomous governance equivalent of SOC 2 in enterprise procurement and institutional due diligence workflows.

Framework	Relationship to ARAF
EIS-01	EIS-01 defines the minimum evidence record structure (four components) required for Tier 1 evidence under this Standard. D6 assessment relies on the two-layer evidentiary model specified in EIS-01 Section 5. Governance agent outputs are governed by EIS-01 Section 7.

Annexure C — Citation and Reliance Guidance

This annexure is informative. It provides guidance for institutional audiences on how to cite and rely on ARAF assessments.

C.1 Canonical Citation

Martin, Carly. Agentic Risk Architecture Framework (ARAF) Standard, Version 3.0. Institute for Autonomous Governance Pty Ltd, 2026.

C.2 Reliance Boundaries

An ARAF certification report is a governance assessment conducted at a point in time. Institutional audiences relying on a certification should observe the following:

- Reliance Freshness: a certification report may be relied upon for 60 days from its issue date without reassessment. After 60 days, institutional audiences should confirm that no triggering events have occurred and that reassessment has not been initiated.
- Scope: certification covers the governance architecture of the autonomous system deployment as scoped in the assessment. It does not cover other systems, business units, or deployment contexts.
- Evidence tier: the evidence tier used per dimension affects the confidence warranted in the assessment. Tier 1 evidence supports higher confidence than Tier 3 reconstruction.
- Limitations: certification does not guarantee future system performance, safety, or absence of adverse outcomes. It is contemporaneous evidence of governance posture at assessment, not a guarantee of future conduct.

C.3 Institutional Use Guidance

Boards: require ARAF Assessed status as a minimum before approving production deployment of a material autonomous system. Require multiplier status to be included in governance reports. Request reassessment upon triggering events.

Insurers: use the GBI dimensional profile as the risk tier signal for AE3 underwriting. D4 maps to autonomous action consequences exposure. D1 maps to commitment authority exposure. D6 maps to governance maintenance risk across the coverage period.

Investors: GBI above 2.50 is a remediation liability. GBI below 1.75 is a governance asset. Use the dimensional profile to distinguish structural risks (D3, D4) from operational risks (D1, D6).

Regulators: request ARAF assessments as a comparable, assessable governance signal. The GBI dimensional profile maps to the requirements of the EU AI Act, NIST AI RMF, ISO 42001, APRA CPS 230, and the Corporations Act section 180(1) duty of care standard.

© 2026 Institute for Autonomous Governance Pty Ltd (ACN 696 112 277) | CC BY 4.0 | araf-standard.org